Name: Bhavesh Punjabi
Class: D15A
Roll. No: 40

Experiment 03: Exploratory Data Analysis and Visualization

Aim:

To perform exploratory data analysis (EDA) and data visualization on the movie ticket booking system using Python.

Theory:

Dataset Overview:

The dataset used in this study represents a **Movie Ticket Booking System**. It contains information about customer bookings and movie-related transactions. The dataset includes attributes such as **Booking_ID, Movie, City, Tickets, Ticket_Price, Total_Amount, and Booking_Rating**. Exploratory Data Analysis (EDA) is performed on the raw dataset to understand its characteristics, distributions, relationships, and patterns before applying further analysis or predictive techniques.

Significance of EDA for movie ticket booking :

EDA is an important step in data analysis because it helps understand the quality and behavior of the movie ticket booking data.

1. Data Cleaning and Quality Check
EDA helps identify missing values, inconsistent data and unusual observations in the booking dataset. For example, unusually high booking amounts or ticket quantities can be detected during analysis.

2. Understanding Data Distribution
Statistical measures and visualizations help understand how variables such as ticket price, number of

tickets and total booking amount are distributed. This can help identify whether the data contains skewness or unusual values.

3. Identifying Important Features

EDA helps identify attributes that are useful for understanding customer booking behavior. For example, **Movie, City, Tickets, Ticket_Price, and Total_Amount** can provide important information about booking patterns.

4. Detecting Relationships and Patterns

Correlation analysis helps identify relationships between numerical attributes. For example, the number of tickets booked is expected to have a positive relationship with the total booking amount.

5. Improving Decision Making

EDA provides useful insights into popular movies, booking quantities, ticket prices and customer spending patterns. These insights can help a movie ticket booking system improve its services and understand customer preferences.

1. Descriptive Analysis - Statistical Measures of Central Tendency

Measures of central tendency describe the center or typical value of a dataset. The main measures are:

- **Mean:** Represents the average value of a numerical attribute, such as the average number of tickets booked.
- **Median:** Represents the middle value when the observations are arranged in order. It is less affected by extreme values.
- **Mode:** Represents the most frequently occurring value. It can be useful for identifying the most common ticket price or booking quantity.

2. Descriptive Analysis - Statistical Measures of Dispersion

Measures of dispersion describe how widely the values are spread around the center of the dataset.

The main measures include:

- **Standard Deviation:** Shows how much the values differ from the mean.
- **Variance:** Measures the spread of data and is the square of standard deviation.

- **Range:** Represents the difference between the maximum and minimum values.
- **Interquartile Range (IQR):** Represents the range of the middle 50% of observations and can be useful for detecting outliers.

3. Correlation Between Attributes

Correlation measures the strength and direction of the relationship between numerical variables. The **Pearson correlation coefficient** ranges from **-1 to +1**.

For example:

- A positive correlation between **Tickets** and **Total_Amount** indicates that customers booking more tickets generally have a higher total booking amount.
- The relationship between **Ticket_Price** and **Total_Amount** can also be studied to understand their effect on total spending.
- Weak correlation between two attributes indicates that they have a limited linear relationship.

Correlation analysis helps identify important relationships and patterns within the movie ticket booking data.

4. Different Visualization Techniques & Their Uses

Visualization makes it easier to understand patterns, distributions and relationships in the dataset.

- Histogram: Shows the distribution of numerical data such as ticket prices.
- Boxplot: Shows the spread of data and helps identify possible outliers in total booking amounts.
- Heatmap: Displays the correlation between numerical attributes using different intensity levels.
- Scatter Plot: Shows the relationship between two numerical variables, such as the number of tickets and total booking amount.
- Bar Chart: Compares categorical data, such as the number of bookings for different movies or cities.

Code:

Step 1: Loading the Dataset and Initial Exploration Output:

```
data = {
    'Booking_ID': range(1, 21),
    'Movie': [
        'Avengers', 'Avatar', 'Jawan', 'Pathaan', 'Leo',
        'KGF 2', 'Animal', 'Dangal', 'RRR', 'Pushpa',
        'Jawan', 'Avatar', 'Leo', 'Animal', 'RRR',
        'KGF 2', 'Dangal', 'Pushpa', 'Pathaan', 'Avengers'
    ],
    'City': [
        'Mumbai', 'Pune', 'Mumbai', 'Delhi', 'Pune',
        'Mumbai', 'Delhi', 'Pune', 'Mumbai', 'Delhi',
        'Pune', 'Mumbai', 'Delhi', 'Mumbai', 'Pune',
        'Delhi', 'Mumbai', 'Pune', 'Mumbai', 'Delhi'
    ],
    'Tickets': [2, 4, 3, 5, 2, 6, 3, 4, 5, 2,
                3, 5, 2, 4, 6, 3, 5, 2, 4, 3],
    'Ticket_Price': [250, 300, 220, 250, 280, 200, 300, 220, 250, 200,
                     220, 300, 280, 300, 250, 200, 220, 200, 250, 250],
    'Total_Amount': [500, 1200, 660, 1250, 560, 1200, 900, 880, 1250, 400,
                     660, 1500, 560, 1200, 1500, 600, 1100, 400, 1000, 750],
    'Booking_Rating': [5, 4, 5, 4, 3, 5, 4, 4, 5, 3,
                       5, 4, 3, 5, 4, 4, 5, 3, 4, 5]
}

df = pd.DataFrame(data)

df.head()
```

	Booking_ID	Movie	City	Tickets	Ticket_Price	Total_Amount	Booking_Rating
0	1	Avengers	Mumbai	2	250	500	5
1	2	Avatar	Pune	4	300	1200	4
2	3	Jawan	Mumbai	3	220	660	5
3	4	Pathaan	Delhi	5	250	1250	4
4	5	Leo	Pune	2	280	560	3

First 5 records:

	Booking_ID	Movie	City	Tickets	Ticket_Price	Total_Amount	Booking_Rating
0	1	Avengers	Mumbai	2	250	500	5
1	2	Avatar	Pune	4	300	1200	4
2	3	Jawan	Mumbai	3	220	660	5
3	4	Pathaan	Delhi	5	250	1250	4
4	5	Leo	Pune	2	280	560	3

```
Dataset Shape:
(20, 7)

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20 entries, 0 to 19
Data columns (total 7 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   Booking_ID      20 non-null     int64
 1   Movie           20 non-null     object
 2   City            20 non-null     object
 3   Tickets         20 non-null     int64
 4   Ticket_Price    20 non-null     int64
 5   Total_Amount    20 non-null     int64
 6   Booking_Rating  20 non-null     int64
dtypes: int64(5), object(2)
memory usage: 1.2+ KB
```

Statistical Summary:

	Booking_ID	Tickets	Ticket_Price	Total_Amount	Booking_Rating
count	20.00000	20.000000	20.000000	20.000000	20.000000
mean	10.50000	3.650000	247.000000	903.500000	4.200000
std	5.91608	1.348488	36.433068	355.709447	0.767772
min	1.00000	2.000000	200.000000	400.000000	3.000000
25%	5.75000	2.750000	220.000000	590.000000	4.000000
50%	10.50000	3.500000	250.000000	890.000000	4.000000
75%	15.25000	5.000000	280.000000	1200.000000	5.000000
max	20.00000	6.000000	300.000000	1500.000000	5.000000

Step 2: Task 1 - Descriptive Analysis - Statistical Measures of Central Tendency

[5]
✓ 0s

```python
print("Mean Tickets:", df['Tickets'].mean())
print("Median Tickets:", df['Tickets'].median())
print("Mode Tickets:", df['Tickets'].mode()[0])

print("\nMean Ticket Price:", df['Ticket_Price'].mean())
print("Median Ticket Price:", df['Ticket_Price'].median())
print("Mode Ticket Price:", df['Ticket_Price'].mode()[0])

print("\nMean Total Amount:", df['Total_Amount'].mean())
print("Median Total Amount:", df['Total_Amount'].median())
print("Mode Total Amount:", df['Total_Amount'].mode()[0])
```

```
Mean Tickets: 3.65
Median Tickets: 3.5
Mode Tickets: 2

Mean Ticket Price: 247.0
Median Ticket Price: 250.0
Mode Ticket Price: 250

Mean Total Amount: 903.5
Median Total Amount: 890.0
Mode Total Amount: 1200
```

Inferences:

- The mean represents the average number of tickets booked, ticket price and total booking amount.
- The median represents the middle value and is less affected by unusually large bookings.
- The mode identifies the most frequently occurring ticket quantity and ticket price.
- These measures help identify the typical booking behavior of customers.

Step 3: Task 2 - Descriptive Analysis - Statistical Measures of Dispersion

```
columns = ['Tickets', 'Ticket_Price', 'Total_Amount']

for col in columns:
    print("\n", col)
    print("Standard Deviation:", df[col].std())
    print("Variance:", df[col].var())
    print("Range:", df[col].max() - df[col].min())
    print("IQR:", df[col].quantile(0.75) - df[col].quantile(0.25))
```

```
 Tickets
Standard Deviation: 1.3484884325167863
Variance: 1.8184210526315792
Range: 4
IQR: 2.25

 Ticket_Price
Standard Deviation: 36.4330676865486
Variance: 1327.3684210526317
Range: 100
IQR: 60.0

 Total_Amount
Standard Deviation: 355.7094467768825
Variance: 126529.21052631579
Range: 1100
IQR: 610.0
```

Inferences:

- Standard deviation shows the variation in the number of tickets, ticket price and total booking amount.
- Variance provides another measure of how widely the booking values are distributed.
- Range shows the difference between the highest and lowest booking values.
- IQR helps understand the spread of the middle 50% of bookings and identify possible outliers.

Step 4: Task 3 - Correlation Between Attributes

```
numeric_df = df[['Tickets', 'Ticket_Price', 'Total_Amount', 'Booking_Rating']]

correlation = numeric_df.corr()

print("Correlation Matrix:")
display(correlation)
```

```
Correlation Matrix:
```

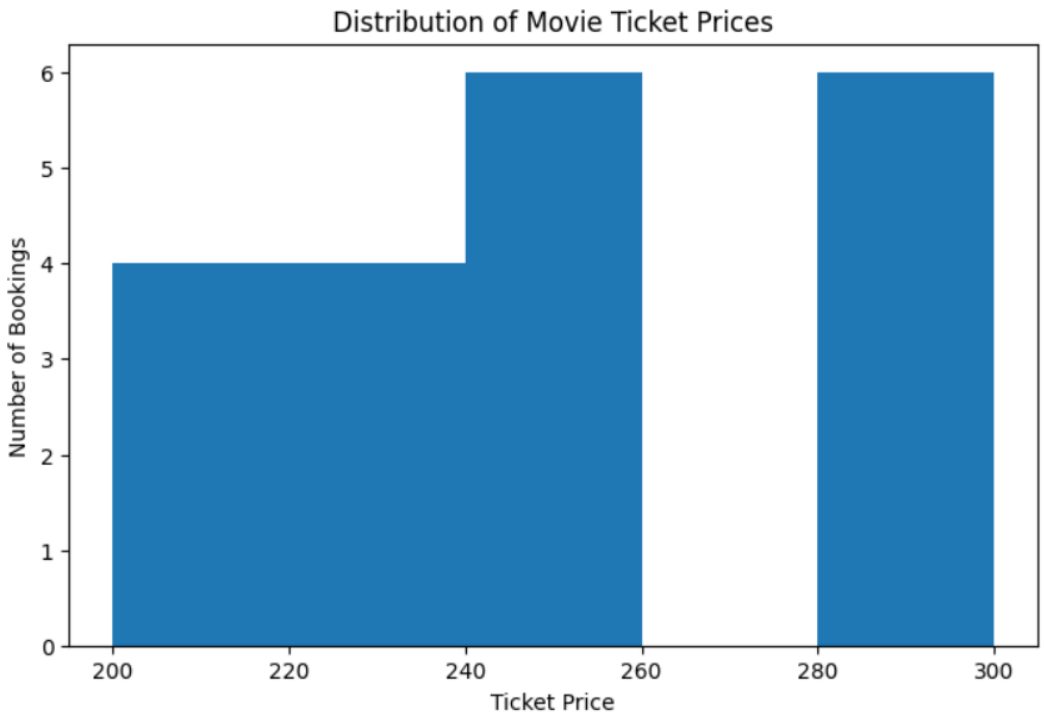
	Tickets	Ticket_Price	Total_Amount	Booking_Rating
Tickets	1.000000	0.041780	0.923277	0.427018
Ticket_Price	0.041780	1.000000	0.401288	-0.071499
Total_Amount	0.923277	0.401288	1.000000	0.344192
Booking_Rating	0.427018	-0.071499	0.344192	1.000000

Next steps: Generate code with correlation | New interactive sheet

Inferences:

- **Tickets and Total_Amount** are expected to have a strong positive relationship because booking more tickets generally increases the total amount.
- **Ticket_Price and Total_Amount** can also show a positive relationship.
- Correlation between **Booking_Rating** and other numerical attributes can help determine whether customer ratings vary with booking behavior.
- Weak correlations indicate that the corresponding attributes have limited linear relationships.

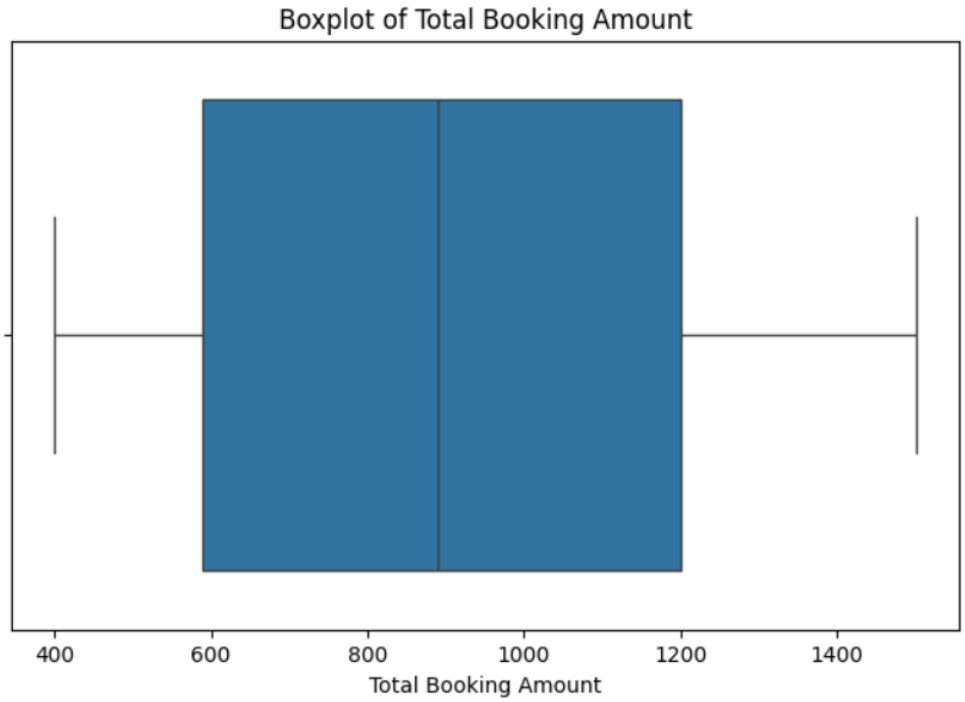
Step 5 : Histogram



Inferences:

- The histogram shows the distribution of ticket prices in the dataset.
- It helps identify the most common ticket-price range.
- It also makes it easier to observe whether ticket prices are evenly distributed or concentrated around certain values.

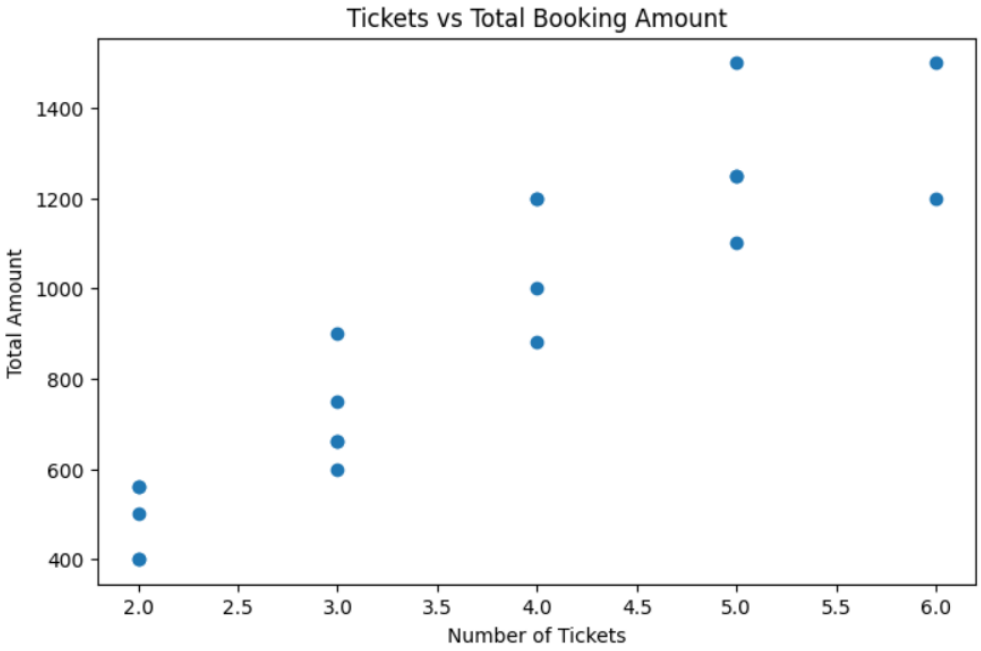
Step 6: Boxplot



Inference:

- The boxplot shows the spread of Total_Amount values.
- It helps identify the median and variation in booking amounts.
- Any values appearing far from the main distribution may indicate unusually large or small bookings**.**

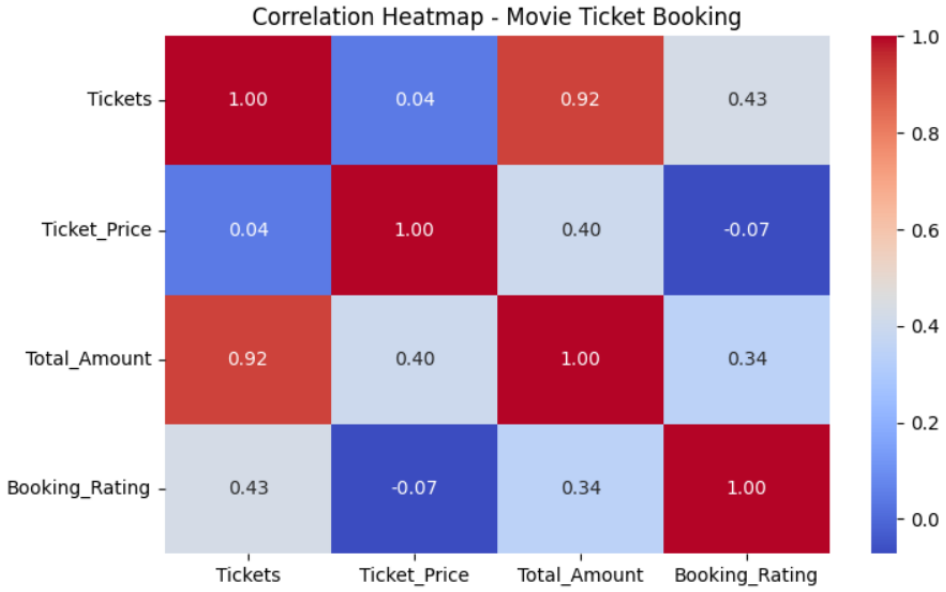
Step 7 : scatter plot



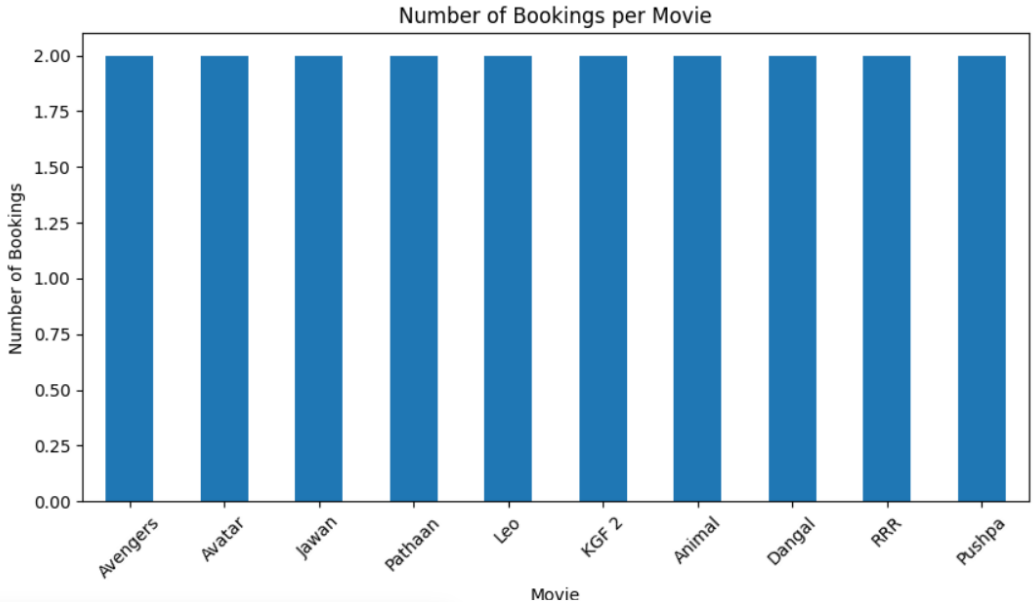
Inferences :

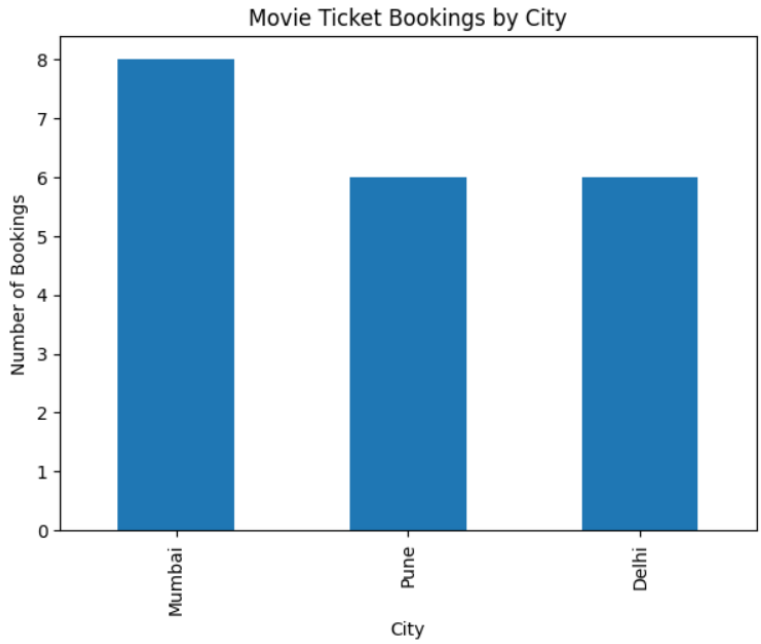
- The scatter plot shows the relationship between Tickets and Total_Amount.
- A positive upward trend indicates that customers booking more tickets generally have higher total booking amounts.
- The plot also helps identify unusual booking combinations.

Step 8 : Heatmap



Step 9: Movie & City wise booking analysis





Conclusion:

The exploratory data analysis of the Movie Ticket Booking dataset provided useful insights into customer booking behavior. Descriptive statistics helped understand the typical number of tickets, ticket prices and total booking amounts. Dispersion measures showed the variation in customer spending. Correlation analysis identified the relationship between the number of tickets and total booking amount. Histograms, boxplots, scatter plots, heatmaps and bar charts helped visualize ticket prices, booking amounts, relationships and movie popularity. Overall, EDA helps the movie ticket booking system understand customer preferences, popular movies and booking patterns, which can support better decision-making and service improvement.

==========

GitHub Repository Link:

https://github.com/Sentry2311/DMBI_EXPERIMENTS/tree/main/Experiment_03_Exploratory_Data_Analysis_Visualization

(Contains complete documentation, dataset, and program code for Experiment 3: Exploratory Data Analysis and Visualization)